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SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

CYBER SECURITY INCIDENT RESPONSE MANAGEMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Your organization receives a call from a federal law enforcement agency, informing you that they have information indicating a data breach occurred involving your environment. The agency provides a number of specific details, including the date and time when sensitive data was transferred out of your network, the IP address of the destination, and the nature of the content. **10**

i. Does this information match the characteristics of a good lead? Explain why or why not.

ii. What else might you ask for?

iii. How can you turn this information into an actionable lead?

b. Discuss some of the challenges related to outsourced IT services as they pertain to conducting an investigation. How can they be dealt with? **04**

2. a. Assume that a security model is needed for the protection of information in your class. Using the CNSS model, examine each of the cells and write a brief statement on how you would address the three components occupying that cell. **04**

b. **Case** - The next day at SLS found everyone in technical support busy restoring computer systems to their former state and installing new virus and worm control software. Amy found herself learning how to install desktop computer operating systems and applications as SLS made a heroic effort to recover from the attack of the previous day. **10**

i. Do you think this event was caused by an insider or outsider? Why do you think this?

ii. Other than installing virus and worm control software, what can SLS do to prepare for the next incident?

iii. Do you think this attack was the result of a virus or a worm? Why do you think this?

3. a. Assume your web server farm was found to be operating as a "drop site" for attackers who were active in other companies. What would your investigative priorities be? How would you demonstrate that the incident was handled in a competent manner? **07**

b. When maintaining a timeline of attacker activity, what four pieces of information should you record for each entry? Pick a file, such as win.ini on a Microsoft Windows computer or /etc/hosts on a Unix-based host, and make as many time related entries as you can for just that one file. **07**

OR

4. a. **Case** - The next morning at 8 o'clock, Kelvin called the meeting to order. The first person to address the group was the network design consultant, Susan Hamir. She reviewed the critical points from her earlier design report, going over the options it had presented and outlining the trade-offs in those design choices. When she finished, she sat down and Kelvin addressed the group again: "We need to break the log jam on this design issue. We have all the right people in this room to make the right choice for the company. Now here are the questions I want us to consider over the next three hours." Kelvin pressed the key on his PC to show a slide with a list of discussion questions on the projector screen. **10**

   i. What questions do you think Kelvin should have included on his slide to start the discussion?

   ii. If the questions to be answered were broken down into two categories, they would be cost versus maintaining high security while keeping flexibility. Which is most important for SLS?

b. What are the differences between a policy, a standard, and a practice? What are the three types of security policies? Where would each be used? What type of policy would be needed to guide use of the Web? E-mail? Office equipment for personal use? **04**

5. a. Go to a popular online electronic commerce site like Amazon.com. Place several items in your shopping cart, and then go to check out. When you reach the screen that asks for your credit card number, right-click on the Web browser and select "Properties." What can you find out about the cryptosystems and protocols in use to protect this transaction? **10**

b. How does a false positive alarm differ from a false negative one? From a security perspective, which is least desirable? **04**

OR

6. a. Charlie was just getting ready to head home when the phone rang. Caller ID showed it was Peter. **07**

"Hi, Peter," he said into the receiver. "Want me to start the file cracker on your spreadsheet?"

"No, thanks," Peter answered, taking the joke well. "I remembered my passphrase. But I want to get your advice on what we need to do to make the use of encryption more effective and to get it properly licensed for the whole company. I see the value in using it for certain kinds of information, but I'm

worried about forgetting a passphrase again or even worse, that someone else forgets a passphrase or leaves the company. How would we get their files back?"

"We need to use a feature called key recovery, which is usually part of PKI software," said Charlie. "Actually, if we invest in PKI software, we could solve that problem as well as several others."

"OK," said Peter. "Can you see me tomorrow at 10 o'clock to talk about this PKI solution and how we can make better use of encryption?"

i. Was Charlie exaggerating when he gave Peter an estimate for the time that would be required to crack the encryption key using a brute force attack?

ii. Are there any tools that someone like Peter can use safely, other than key recovery, to avoid losing his or her passphrase?

b. **Case** - Charlie looked across his desk at Kelvin, who was absorbed in the sheaf of handwritten notes from the meeting. Charlie had asked Kelvin to come his office to discuss the change control meeting that had occurred earlier that day. **07**

"So what do you think?" he asked.

"I think I was blindsided by a bus!" Kelvin replied.

"I thought I had considered all the possible effects of the change in my project plan. I tried to explain this, but everyone acted as if I had threatened their jobs."

"In a way you did," Charlie stated. "Some people believe that change is the enemy."

"But these changes are important."

"I agree," Charlie said. "But successful change usually occurs in small steps. What's your top priority?"

"All the items on this list are top priorities," Kelvin said. "I haven 't even gotten to the second tier"

"So what should you do to accomplish these top priorities?" Charlie asked.

"I guess I should reprioritize within my top tier, but what then?"

"The next step is to build support before the meeting, not during it." Charlie smiled. "Never go into a meeting where you haven't done your homework, especially when other people in the meeting can reduce your chance of success."

i. What project management tasks should Kelvin perform before his next meeting?

ii. What change management tasks should Kelvin perform before his next meeting, and how do these tasks fit within the project management process?

iii. Had you been in Kelvin's place, what would you have done differently to prepare for this meeting?

7. a. In an investigation, a network administrator notifies you that a Mac OS X system appears to be generating significant traffic that doesn't correlate with normal activity. How would you identify the source of the traffic on the system? **07**

b. You are investigating what artifacts an application creates on a Windows 8 system. You notice that under the AppData directory, the application is storing information in multiple directories: Local, LocalLow, and Roaming. Why would the application do that? What is the purpose of the different AppData directories (Local, LocalLow, and Roaming)? **07**

OR

8. a. Assume that you have been asked to review the power needs for a standalone computer system which processes important but noncritical data and does not have to be online at all times, and which stores valuable data that could be corrupted if the power to the system were suddenly interrupted. Which UPS features are most important to such a system? Which type of UPS do you recommend for this system? **07**

b. Define the required wattage for a UPS for the following systems: **07**

i. Monitor: 2 amps; CPU: 3 amps; printer: 3 amps

ii. Monitor: 3 amps; CPU: 4 amps; printer: 3 amps

iii. Monitor: 3 amps; CPU: 4 amps; printer: 4 amps

Search the Web for a UPS that provides the wattage necessary to run the systems listed above for at least 15 minutes during a power outage.